

Shadow XR Visual Paradigm — Design Rationale

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Version: 1.0 · **Prepared:** 2026-07-06 (late night, post-full-web-deep-research) **Supersedes:** the “5 personas floating in semicircle + 4 Flow demos + 3D Gaussian Splatting ambient” plan. **Authority:** cross-referenced against Citi × 8ninths (2016), Fidelity StockCity postmortem (2014), Kraus et al. CGF 2022 immersive-analytics survey, Cordeil ImAxes/IATK, Redd 2025 chemistry-VR replication, Flow Immersive published gallery, and 2026 Big-4 AI audit market scans.

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What we are NOT building

Every publicly-shipped bank AR/VR demo from 2014 through 2026 has died in the same trap: **impressive for 30 seconds, useless at day 30**. Named case studies:

- **Fidelity StockCity (2014)** — Oculus SimCity metaphor for portfolios. Never shipped consumer.
- **Citi Holographic Workstation (2016)** — HoloLens 3-tier trading dockspace. Piloted, never rolled out.
- **JPMorgan Onyx (2022, Decentraland)** — branded lobby. Not a workflow tool.
- **HSBC × The Sandbox (2022)** — 3×3 metaverse LAND for gaming engagement. Marketing.
- **Meta Horizon Workrooms** — discontinued 2026-02-16 after enterprise failure.
- **Wells Fargo Together Experience** — Oculus + treadmill mall tour. Novelty.

Cargo-cult 3D pie charts, floating city metaphors, and novelty spatial layouts all fail the day-30 utility test. **Do not build any of these.**

What we ARE building

Two academic + industry evidence categories replicate under expert scrutiny:

1. **Spatial arrangement of familiar 2D artifacts** (Flow Immersive portfolio panels, Citi 3-tier dockspace, Tableau visionOS floating dashboards)
2. **Embodied outlier / cluster identification** in 3D scatter (Kraus et al. CGF 2022, Filho et al. replicated by Redd 2025 in *J. Chemometrics*)

Shadow's design combines both plus one specific new claim:

The only 3D element that earns its 3D existence is one that visualizes a *relationship* the 2D UI cannot afford.

For Shadow, that relationship is **cryptographic audit provenance** — the hash chain linking every decision back through the reason-code dictionary hash to the model version. This has no clean 2D representation because it is time-ordered + graph-structured + signature-bound. Spatializing it gives a bank auditor exactly the “walk the chain to inspect a specific link” affordance the 2D UI can’t produce. Everything else stays flat.

The design: Semicircular Tribunal

Geometry

- Reviewer stands at origin (0, 1.6, 1) looking toward -Z. Room extent 4×3×4 meters.
- **Five persona cards** float in a semicircle at radius 1.5m, height 1.6m (eye level), spanning -60° to +60° in the reviewer’s forward 120° cone.
- Each card is a **flat 2D panel** (approximately 30cm wide × 45cm tall in world space, matching a large iPad).
- **Central hash-chain ribbon** — the only true 3D element — rises vertically at (0, 0.8m to 2.4m, -1.0m). Each link of the ribbon corresponds to one prior attestation (24-hour rolling window). Reviewer walks toward the ribbon to inspect individual links; walks around to see the chain from side view; walks through if the room permits.

Persona card content

Each card shows the same fields as `/api/deliberate` returns per voice:

- **Verdict pill** — colored green (approve), amber (escalate), red (block)
- **Voice name** in header
- **Rationale short** — one paragraph
- **AA code** if applicable
- **Confidence weight** — small numeric badge in the corner
- **Signature footer** — abbreviated attestation hash (first 8 chars)

Layout is intentionally the same as a printed regulator submission page. Familiarity is the point. Bank counsel opens the demo and reads the same layout they would read on paper.

Hash-chain ribbon (the only 3D element)

- Vertical column of horizontal ring-segments, one per attestation.
- Rings colored to indicate signature-verification status: **white/silver = verified, red = tampered/broken**.
- Ring label shows the attestation timestamp and short signature hash.
- Below each ring, a **branch** trails outward if that decision has an outstanding compliance flag (AML/KYC or fair-lending review). Reviewer follows the branch to a callout showing the flag context.

The chain is the visual signature of Shadow. It is what the demo is *about*.

What is deliberately NOT in the scene

- No floating 3D scatter plots of loan data (that goes in a *separate* Flow Immersive Brier calibration demo, targeted at the Columbia statistics faculty audience, not the Dean + VP).
 - No ambient 3D Gaussian Splatting background (too heavy for phone-tethered XREAL One Pro; adds no signal).
 - No animated persona avatars, gestures, or voice-over synthesis. Voice output is via the reviewer's phone speakers only.
 - No stock-city / dashboard-city / any -city metaphor.
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The hardware pipeline

Correction to the 2026-07-06 morning test protocol

XREAL One Pro is a display peripheral, not a standalone computer. It requires USB-C DisplayPort-Alt-Mode from an iPhone 15+/16/17 or Android with DP-out or Steam Deck / ROG Ally / Mac / PC. The X1 chip handles pose prediction, display stabilization, distortion correction; **it does not run WebGL / WebXR / a browser.**

Correction to Phase B test sessions: standalone rendering is impossible. The pipeline is:

- **Android phone + Chrome + WebXR + Three.js** → HDMI-alt-mode → XREAL One Pro
- Fallback: **Mac + Chrome window (non-WebXR)** cast to XREAL One Pro as a 171" 2D virtual screen. Loses head-tracked rendering; retains the persona-tribunal layout as a fixed image.

iPhone Safari does not support WebXR immersive-AR mode. Do not build for iOS Safari.

Rendering budget on Android + Chrome tethered

- Three.js scene graph, WebGPU where available (WebGL2 fallback)
- Under 100 draw calls total
- Under 100k triangles total (5 persona cards × ~4k tris each + hash-chain 20k tris + reviewer UI 15k tris = ~55k)
- No 3D Gaussian Splatting (mobile CPU-sort drops FPS to 20-30)
- Target 60 fps on Pixel 9 Pro / Samsung S24 Ultra / OnePlus 12
- Fallback path: reduce persona count from 5 to 3, hide chain ring branches, drop to 30 fps target

What to buy in the next 5 days

- **Android host phone** — Pixel 9 Pro (\$999) or borrow a Samsung S24. Must have USB-C with DisplayPort-Alt-Mode. **Do not use iPhone as demo host** (Safari WebXR gap).
- **USB-C to XREAL One Pro cable** — included with the XREAL purchase, verify Loredana confirms it's in the box before Saturday.
- Optional: **compact Bluetooth controller** for reviewer hand-input if MediaPipe Hands is not shipped in the demo.

What ships when

- **2026-07-07 (Tue)**: rebuild the WebXR scene against the tribunal + hash-chain paradigm (delete the earlier semicircle prototype). Target: Chrome DevTools WebXR emulator renders the scene at 60 fps on a Mac.
- **2026-07-08 (Wed)**: package the scene as a Flow Immersive-compatible URL. Upload to `a.flow.gl`. Get share URL.
- **2026-07-11 (Fri/Sat)**: XREAL One Pro arrives at Loredana. Both authors confirm the box includes the DP-alt USB-C cable.
- **2026-07-13 (Sun)**: Council-for-Slack submit. XREAL demo is not on the critical path this weekend.
- **2026-07-14 (Mon)** onward: 5-session on-device test week per `docs/xreal-one-pro-test-protocol/` (corrected version to reflect Android tethering).
- **Late July** (executive Dean + VP demo): live demo with the corrected scene, corrected hardware pipeline, corrected fallback plan.

Roadmap after the July demo

Late 2026: XREAL Aura ships (\$1,500, Fall 2026 per XREAL's Project Aura page). Aura is standalone (Android XR compute puck), which lets us drop the phone-tethering constraint. Migration cost: minimal, since the scene is already WebXR + Three.js — the Aura runs the same Chromium-based Android browser.

2027: Anthropic + Apple + Meta smart-glasses landings likely reshape the market. Track the space, but do not build the pitch around any of it. The pitch is that Shadow runs on **commodity AR glasses banks can buy at Best Buy today for \$599**.

Publications: this design directly feeds the IEEE VR 2027 paper (`Correspondence/ECC-2026/ieee-vr-2027-abstract-v3-2026-07-06.md`). Update the Method section (v2 Section 4.5 "Spatial XR layer") to reference this document as the canonical design source.

Reviewer notes for Loredana

1. This document changes the demo from "5 personas floating with 3D scatter ambient" to a much more sober "tribunal of 2D verdict cards + one true 3D hash-chain ribbon." I made the change because the full-web deep-research on 2014–2026 bank AR/VR shipping products showed that every novelty-metaphor demo died in the "day 30" utility test. Sober + defensible + academic-evidence-backed is the right posture for the Dean + VP audience.
2. Please confirm the persona card fields (verdict, rationale, AA code, confidence, signature footer) match what you would want a bank counsel to see first. If any field should be dropped or added, please annotate.

3. The XREAL One Pro purchase is still the right call. It is \$599 hardware banks can actually adopt without a procurement cycle. The Aura roadmap (\$1,500, Fall 2026) is the next-generation story to include in the pitch.
 4. The Android phone requirement is a new constraint I did not flag before Saturday's purchase. If you don't have a Pixel or a Samsung S24 on hand, I can borrow one for the demo week.
 5. I will update `docs/xreal-one-pro-test-protocol/README.md` to reflect the tethered rendering pipeline. Phase B sessions become "tethered baseline" instead of "standalone baseline".
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